

Qatar-1b

R-2832

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130713 · created 2026-09-05T23:41:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456486.867426 +/- 7.9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.36 +/- 1.94 [%]
Orbital Inclination (inc)	79.51 +/- 2.94
Airmass coefficient 1 (a1)	1.15 +/- 0.13
Airmass coefficient 2 (a2)	-0.11 +/- 0.45
Scatter in the residuals of the lightcurve fit is	11.02 %
Best Comparison Star	#2 - [528, 239]
Optimal Aperture	1.36
Optimal Annulus	6.27
Transit Duration (day)	0.028 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.183 $\pm$ 0.053 vs 0.1463 (deviation 0.59 σ)
Duration fit vs published	0.028 d vs 0.0692 d (deviation 1.21 σ)
Mid-time O-C	-0.1 min
Depth SNR	2.9
Residual scatter	11.02 %

Light curve

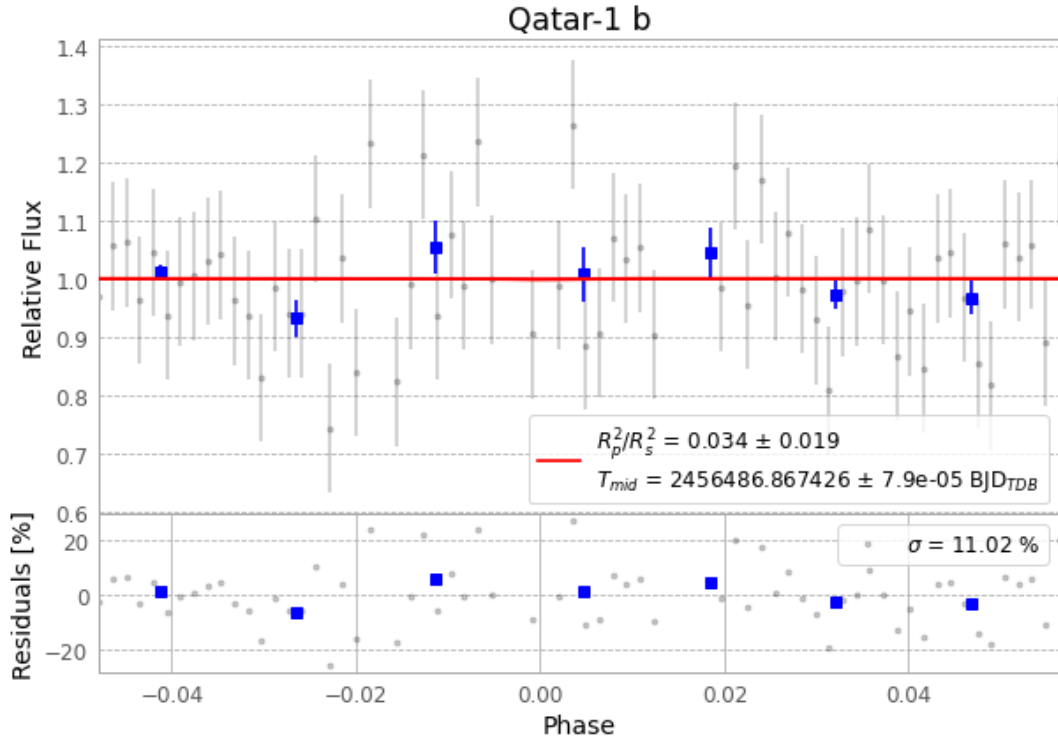


Figure 1 — FinalLightCurve_Qatar-1 b_2013-07-13.png (SHA-256 202a782de1a09b82...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [481, 295], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 86155718648cefb3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1130713070912.FITS → Qatar-1130713104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130713.log (0c6cb6611e3d...), FinalLightCurve_Qatar-1 b_2013-07-13.png (202a782de1a0...), FinalLightCurve_Qatar-1 b_2013-07-13.csv (3caac7ae9d4b...)

Compute resources

Wall time 237.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:41Z.